FIT5196 DATA WRANGLING

Unit Introduction

Faculty of Information Technology

Monash University

Unit Introduction

- Unit Objectives
- Unit Schedule
- Class Timetable
- Teaching Team
- Communication Channels
- Assessments

MONASH University

Unit Objectives

- FIT5196 is trying to achieve
  - parse data in the required format;
  - assess the quality of data for problem identification;
  - resolve data quality issues ready for the data analysis process;
  - integrate data sources for data enrichment;
  - document the wrangling process for professional reporting;
  - write program scripts for data wrangling processes.

- We **DO NOT** focus on
  - Introduction to Python programming, e.g., how to program in Python

You **NEED** be very familiar with Python and the usage of Python packages!

Unit Schedule

Week	Topic
1	Introduction to Data Wrangling
2	Data Wrangling Process & Tasks
3	Regular Expression
4	Exploratory Data Analysis (EDA) Data Discovery & Collection
5	Data Discovery & Collection
6	Data Structuring
7	Data Quality & Anomalies
8	Data Cleaning
9	Data Transformation
10	Data Enrichment
11	Data Validation
12	Advanced Data Wrangling

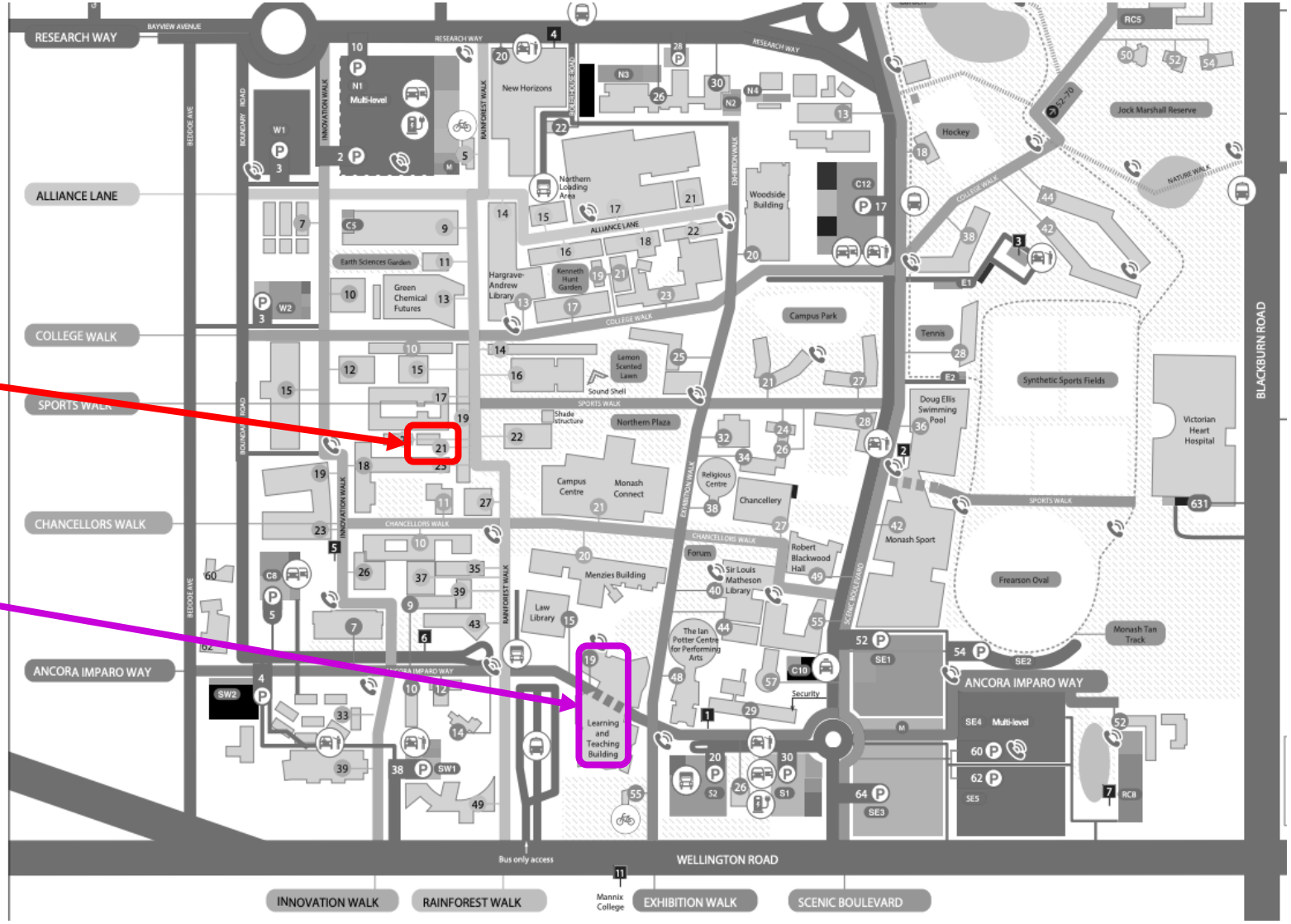
Class Timetable

Every week, you need to attend

- a 2-hour seminar
  - 6pm-8pm Monday
  - S7 Lecture theatre

- a 2-hour applied session
  - LTB, Building
  - Check your class time on Allocate+

Attendance is very important for this unit.



Teaching Team

Jackie Rong
CE, Lecturer

Terrence Mak
Unit Coordinator, Lecturer

Kiara Wang
Head Tutor

Liz Liu
Tutor

Aria Liu
Tutor

Harry He
Tutor

Communication Channels

- Please use **Ed Forum** for discussion and help each other. The teaching team regularly monitors the forum. Please allow up to **3 business days** for us to answer your questions.

- **Weekly Consultation** (via Zoom)
  - Kiara Wang: 9am – 10am, Wednesday
  - Terrence Mak: 12:30pm – 1:30pm, Friday

Email us **ONLY** when you have any emergency to discuss. Please use **private posts** to share your personal queries on **Ed Forum**.

Assessments

	Weight	Tasks	Due Date	Type
Quiz 1	10%	In-applied Session Quiz (MCQs)	**Week 6**	Individual, in-person
Quiz 2	10%	In-applied Session Quiz (MCQs)	**Week 12**	Individual, in-person
Assessment 1	35%	Coding, Report, Demo Video	**23:55, Thursday, Week 7**	Group
Assessment 2	40%	Coding, Report Presentation	**23:55, Thursday, Week 12** **Monday/Tuesday, Week 15**	Group
In-class Participation	5%	In-class activities	**2-hour seminar time in selected weeks**	Individual

Online Quizzes (10%+ 10%)

- Quiz coverage
  - Quiz 1 (10%) covers Week 1-6
  - Quiz 2 (10%) covers Week 7-12
- Students must successfully complete all MCQs in both quizzes to obtain full marks.
- Each quiz has a strict time limit of **30 minutes**.
- Each quiz contributes **10%** to the final unit grade.
- Individual work, no collaboration, no AI assistance, Safe-exam Browser in-use.
- **Must attend the applied session and complete the quizzes in person**.

Assessment 1 (35%)

Exploratory Data Analysis (EDA)

- Data stored in various structures and formats need to be explored, pre-processed and analysed well to gain a comprehensive understanding.
- Summarise your findings from raw and pre-processed data and identify any potential interesting research problems/tasks in the further machine learning analysis.
- In this assessment, you may require to write Python (Python 3) script to
  - Load and extract data from input file(s),
  - Apply simple pre-processing operations on raw data, i.e. data format/type conversion,
  - Apply EDA with appropriate visualizations
- One submission by a **group of 4-5 people**
- Due time: **23:55, Thursday Week 7**
- **No 2-day short extension** is appliable for this group assessments, only **Special Consideration**

Assessment 2 (40%)

Parsing, Cleansing and Integrating Raw Data

- This assessment addresses the most important but time-consuming steps in data wrangling, i.e., cleansing and integrate data. Students are required to inspect, audit and then identify problems existing in the parsed data, and propose appropriate methods to fix these problems
  - different generic and major data problems could be found in the data might include *Lexical errors*, *Irregularities*, *Violations of the Integrity constraint*, *Duplications*, *Inconsistency* and more.
  - Handle missing values
  - Handle outliers
- One submission by a **group of 4-5 people**
- Report due time: **23:55, Thursday Week 12**
- **All group members** need to present their work **in-person in Week 15** (**2nd week of exam period**)
- Hurdle Applied to presentation
- **No 2-day short extension** is appliable for this group assessments, only **Special Consideration**

In-class Participation (5%)

- In seminars, we will do tasks and answer questions together. You will need to provide your answers via **online quizzes** called "**In-class participation (Week X)**" to record your participation and earn points.
- You need to complete all the tasks/questions to earn 1 point for each participation. Incomplete attempt receives 0 mark, no partial participation will be considered.
- Each in-class participation quiz is **ONLY** open for **two hours during seminar period**. Please try your best to attend the seminars and join the activities.
- No in-class participation in Week 1.
- **No short extension** or **special consideration** is applied on this assessment.

Academic Integrity

AVOID THE PENALTIES- SEEK SUPPORT EARLY

- Learn more about Academic Integrity, plagiarism and collusion at Monash:
  - Academic Integrity policy
  - What is Academic Integrity
  - Academic Integrity online tutorials
  - English Connect academic writing programs
  - Library Learning Skills Advisors - one-on-one consultations.